# 数理分析方法

## 关系、预序与偏好

Lecture 104

原课件逐页保留 · 零基础中文讲解 · 证明训练

**材料与编写说明**

课程原页：张顺明教授课件（用户提供）。参考：Gérard Debreu, Theory of Value (1959)，第一章 Mathematics。原页与本讲义的补充、纠错、例题分开呈现。本文为辅助自学材料，不代表教师官方讲义。

**怎样阅读**

先看本页的中文说明，再对照原页符号；遇到证明，先抄出假设和目标，再逐行说明“这一步用的是什么”。遮住证明重写一遍，最后用反例检查条件是否必要。重复或动画展开页仍逐页保留；说明会指出新增步骤或对应的完整论证位置。

“补充”用于补基础或展开原课件略去的论证；“纠错”指出原页需要修正的地方。课程概览及人物介绍页按其原有用途解释，不虚构数学定理。

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# Contents

零基础读证明：随手查阅 . . . . . 2
1 原课件第 1 页：关系的起点 . . . . . 3
2 原课件第 2 页：知识地图 . . . . . 4
3 原课件第 3 页：关系与例子 . . . . . 5
4 原课件第 4 页：预序与偏序 . . . . . 6
5 原课件第 5 页：严格与无差异 . . . . . 7
6 原课件第 6 页：完全性 . . . . . 8
7 原课件第 7 页：关系分类图 . . . . . 9
8 原课件第 8 页：极大与最大 . . . . . 10
9 原课件第 9 页：上界与确界 . . . . . 11
10 原课件第 10 页：完备性预告 . . . . . 12
11 原课件第 11 页：区间 . . . . . 13
12 原课件第 12 页：乘积序 . . . . . 14
13 原课件第 13 页：单调映射 . . . . . 15
14 原课件第 14 页：有序对的正式定义 . . . . . 16
15 原课件第 15 页：七种性质逐项读 . . . . . 17
16 原课件第 16 页：严格偏好的要求 . . . . . 18
17 原课件第 17 页：负传递的等价形式 . . . . . 19
18 原课件第 18 页：严格偏好定义 . . . . . 20
19 原课件第 19 页：传递与无循环的证明 . . . . . 21
20 原课件第 20 页：从严格定义弱偏好 . . . . . 22
21 原课件第 21 页：三个互斥情形 . . . . . 23
22 原课件第 22 页：命题证明展开 . . . . . 24
23 原课件第 23 页：两种原始描述 . . . . . 25
24 原课件第 24 页：弱偏好定义 . . . . . 26
25 原课件第 25 页：反向构造 . . . . . 27
26 原课件第 26 页：构造的性质 . . . . . 28
27 原课件第 27 页：原页无差异公式纠错 . . . . . 29
28 原课件第 28 页：还原弱关系 . . . . . 30
29 原课件第 29 页：本讲复习 . . . . . 31

## 零基础读证明：随手查阅

**对象的层次。** $x \in A$ 表示一个元素属于集合，$A \subseteq B$ 表示 $A$ 的每个元素都在 $B$ 中。$\{x\}$ 是只含 $x$ 的集合，不是 $x$ 本身。$\varnothing$ 是空集，$\{\varnothing\}$ 则含有一个元素。$\mathbb{R}^N$ 中的点是有 $N$ 个坐标的有序列表。

**量词。** $\forall$ 是“对所有”，$\exists$ 是“至少存在一个”。例如“每个学生有一本书”不要求是同一本；“有一本书供每个学生使用”则要求同一本。证明极限时 $\forall \varepsilon > 0\ \exists n_0\ \forall n \geq n_0$ 的顺序不可换：先指定精度，再找统一起点，最后控制这个起点之后的全部项。

**怎样否定。** 否定“所有 $x$ 满足 $P$”得到“存在 $x$ 不满足 $P$”；否定“存在”得到“所有都不”。否定 $P \Rightarrow Q$ 是 $P$ 真且 $Q$ 假。因此要推翻一个定理，必须满足其全部假设又违反结论；拿不满足假设的例子不能当反例。

**四种基本证明。** (1) 包含 $A \subseteq B$：任取 $x \in A$，用定义得到 $x \in B$。(2) 相等 $A = B$：两个方向分别证明。(3) 等价 $P \iff Q$：证明 $P \Rightarrow Q$ 及反向。(4) 反证：把结论否定加入已知，推出矛盾。逆否命题 $\neg Q \Rightarrow \neg P$ 与 $P \Rightarrow Q$ 等价，“反过来” $Q \Rightarrow P$ 则未必成立。

**数列、子列和收敛。** 数列 $x_n$ 是第 $n$ 个对象；子列 $x_{n_k}$ 的指标必须严格递增。$x_n \to x$ 指任意小的误差最终都能保持。前面有限多项不影响极限。抽子列通常是为了用有界性和有限维 Bolzano–Weierstrass 定理获得极限；抽取之后须把原来所有相关量一起限制到同一子列。

**距离的关键估计。** $\|x\| = (\sum_i x_i^2)^{1/2}$，$\|x - y\|$ 是距离。三角不等式 $\|x - z\| \leq \|x - y\| + \|y - z\|$ 允许经中间点拆误差。两段误差各小于 $\varepsilon/2$，总误差便小于 $\varepsilon$；分成 $m$ 段时可各分配 $\varepsilon/m$。这里的 $\varepsilon$ 不是固定的小数，而是任意正精度。

**做证明题的五行草稿。** 先写完整假设；写准确目标；列出可用定义与定理；逐步注明所用条件；最后检查任意性、存在性和量词顺序。出现“显然”时问：若阅卷者要求展开，我能否补出一个集合等价链、一个不等式或一个明确构造？不知道证明的命题要标为引用，不能把待证结论循环当作理由。

# 1 原课件第 1 页：关系的起点




Theory of Value

An Axiomatic Analysis of Economic Equilibrium

Gerard Debreu (1959)

Professor Shunming Zhang

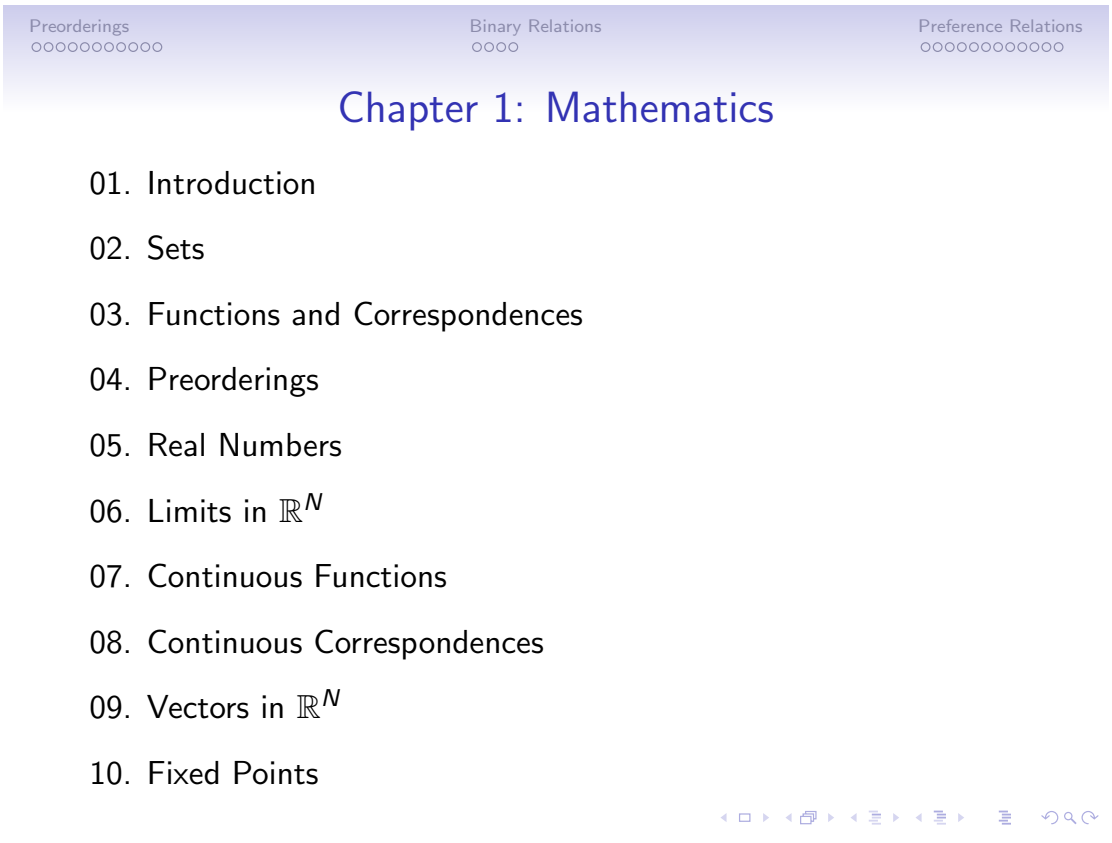
School of Finance
Renmin University of China

September 10, 2022

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

二元关系比较两个对象。函数强调输入输出，关系强调哪些有序对满足条件；一个输入可与多个对象有关。

## 2 原课件第 2 页：知识地图




上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

集合、函数之后引入关系，再用关系描述大小、偏好和确界。实数的顺序公理会用到这一讲的语言。

# 3 原课件第 3 页：关系与例子



## Preorderings

Let $\mathscr{R}$ be a binary relation in which any two elements $x_1$ and $x_2$ of $S$ (the order of which is essential) stand or do not stand. If they do, one writes $x_1 \mathscr{R} x_2$.

Example: (1) Let $\mathscr{S}$ be the set of subsets of a set $S$. The relation $\mathscr{R}$ on $\mathscr{S}$ might be "is contained in," then $X \mathscr{R} X'$ would be equivalent to $X \subseteq X'$: (2) or the relation $\mathscr{R}$ might be "does not intersect," then $X \mathscr{R} X'$ would be equivalent to $X \cap X' = \varnothing$. (3) Let $A$ be the set of actions available to an agent; the relation $\mathscr{R}$ might be "is not preferred to." (4) Consider the relation "is not a successor of" on $\mathbb{N}$ and denote it by $\leqq$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$R \subseteq S \times S$，$xRy$ 是 $(x, y) \in R$ 的简写。包含关系是子集之间的关系，而不是元素之间的属于关系；两者符号不可交换。

# 4 原课件第 4 页：预序与偏序



## Preorderings

The last relation corresponds to the natural ordering of the elements of $\mathbb{N}$. To define with full generality an ordering relation on a set, one preserves certain properties of $\leqq$ on $\mathbb{N}$. In a precise manner, a binary relation $\mathscr{R}$ on $S$ which satisfies

(1) $x\mathscr{R}x$ for every $x \in S$ (reflexivity),

(2) "$x_1\mathscr{R}x_2$ and $x_2\mathscr{R}x_3$" implies "$x_1\mathscr{R}x_3$" (transitivity)

is called a preordering (often also a quasi-ordering). When, in addition, "$x_1\mathscr{R}x_2$ and $x_2\mathscr{R}x_1$" implies "$x_1 = x_2$", the relation is called an ordering. Often the symbols $\precsim$ will be used (in place of $\mathscr{R}$) to denote a preordering. By definition, $x_2 \succsim x_1$ means $x_1 \precsim x_2$.

Example: In the examples of 1.4.a, the first and the fourth relation are preorderings and (in fact, they are orderings), the second is not. The third will always be assumed below to be a preordering: "$a^1$ is not preferred to $a^2$" will be denoted $a^1 \precsim a^2$, this relationship is not necessarily an ordering.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

预序满足自反和传递；偏序额外要求反对称。反对称是“双向关系推出相等”，不是“永远不能双向”。例如通常的 $\leq$ 在 $x = y$ 时允许双向。

# 5 原课件第 5 页：严格与无差异



## Preorderings

"$x_1 \precsim x_2$ and $x_2 \precsim x_1$" is written "$x_1 \sim x_2$". "$x_1 \precsim x_2$ and not $x_2 \precsim x_1$" is written "$x_1 \prec x_2$".

Example: In the last example, $a^1 \sim a^2$ is read "$a^1$ is indifferent to $a^2$" and $a^1 \prec a^2$ is read "$a^1$ is less desired than $a^2$" or "$a^2$ is preferred to $a^1$"

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$x \sim y$ 表示双向弱关系；$x \prec y$ 表示一个方向成立且反方向不成立。无差异未必是对象相同，例如两种不同消费带来同样评价。

# 6 原课件第 6 页：完全性



## Preorderings

Two elements $x_1$ and $x_2$ of a preordered set may not be comparable (as the example of the preordering $\subseteq$ emphasizes). When one necessarily has $x_1 \precsim x_2$ or $x_2 \precsim x_1$ (or both), the preordering is called complete (to emphasize that a certain is not necessarily complete, one calls it partial). Then $x_1 \precsim x_2$ is the negation of $x_1 \succ x_2$.

Example: The preordering "is not preferred to" on the set of actions available to an agent will always be assumed below to be complete.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

完全性要求任意两点可比较。坐标序中 $(1,0)$ 与 $(0,1)$ 不可比较，所以不是完全序。注意“不能证明谁更好”不是数学上的无差异。

# 7 原课件第 7 页：关系分类图



## Preorderings

Figure 1 summarizes the different types of preorderings. An arrow indicates decreasing generality.

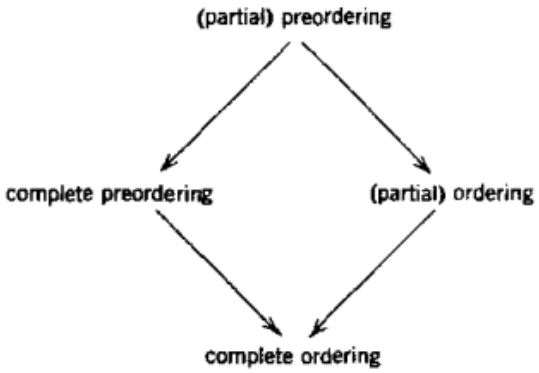



Fig. 1

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沿箭头增加条件会缩小适用对象。检查一种关系属于哪类，依次检查自反、传递、反对称、完全；否定一个性质只需找到违反它的具体对象。

# 8 原课件第 8 页：极大与最大



## Preorderings

Let $S$ be a partially preordered set. When $y \in S$ and there is no $x \in S$ such that $x \succ y$ (resp. $x \prec y$), $y$ is called a maximal (resp. minimal) element of $S$. When $y \in S$ and for all $x \in S$ one has $x \precsim y$ (resp. $x \succsim y$), $y$ is called a greatest (resp. least) element of $S$. When the preordering is complete the converse is also true and the distinction disappears. When the preordering is an ordering there is at most one greateat element and at most one least element.

Example: (1) Consider the set $\mathbb{N}$ partially ordered by the relation "is a divisor of," and let $S$ be $\{1, 2, 3, 4, 5\}$. $S$ has three maximal elements, 3, 4, 5, no greatest element, one minimal element, 1, and one least element, 1. (2) Consider a set $A$ of actions completely preordered by the relation "is not preferred to." If $a$ is an action to which none is preferred, $a$ is a greatest element of $A$ (which need not be unique: there may be other greatest elements, all indifferent to $a$).

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

极大是不被严格压过，最大是压过所有其他对象。整除序里 $3, 4, 5$ 都极大，但没有最大。完全预序中极大即最大：任一点与它可比较，不能严格更大，就只能不超过它。

# 9 原课件第 9 页：上界与确界



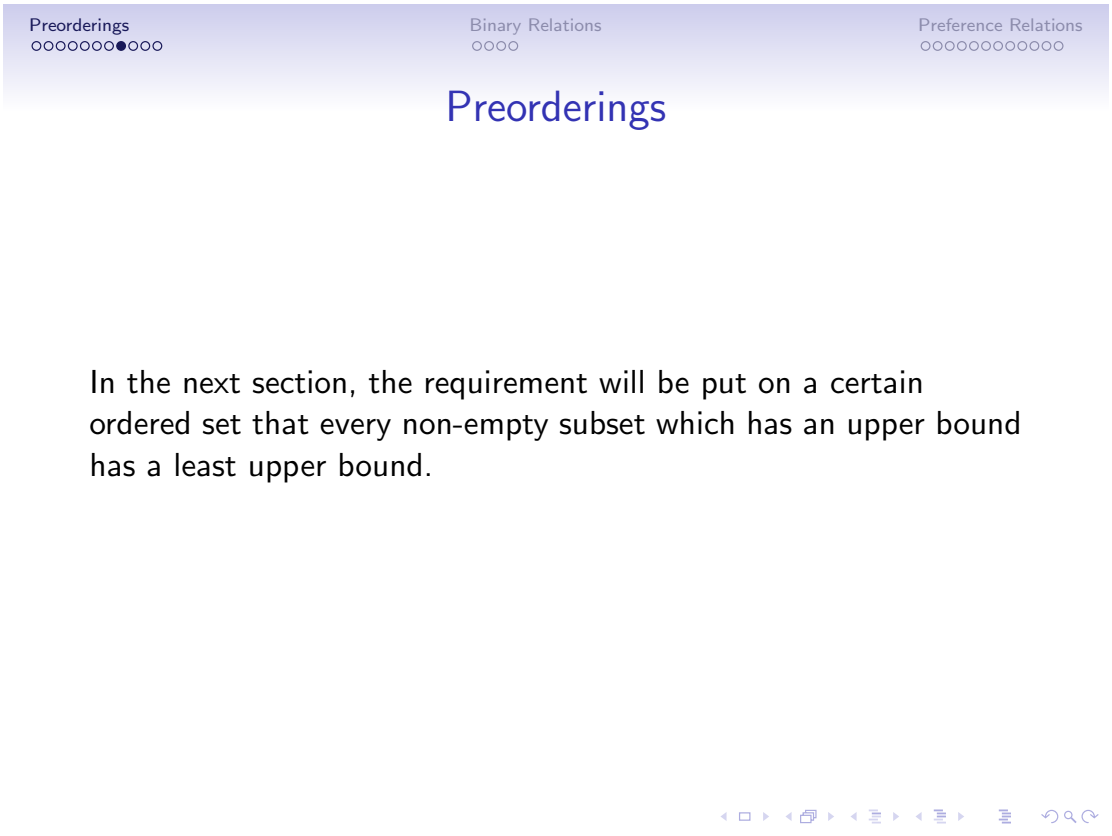
## Preorderings

Let $S$ be a preordered set, and consider a subset $X$ of $S$. An element $y \in S$ such that for all $x \in X$ one has $x \precsim y$ (resp. $x \succsim y$) is called an upper bound of $X$ (resp. a lower bound of $X$). Consider the set $Y$ of the upper bounds of $X$; the set $Y$ has the property: "$y \in Y$ and $y' \succsim y$" implies "$y' \in Y$." A least element of $Y$ is a least upper bound of $X$. If $X$ has a greatest element $y$, $y$ is clearly a least upper bound. Consider the set $Y'$ of the lower bounds of $X$; the set $Y'$ has the property: "$y \in Y'$ and $y \succsim y'$" implies "$y' \in Y'$." A greatest element of $Y'$ is a greatest lower bound of $X$. If $X$ has a least element $y$, $y$ is clearly a greatest lower bound.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$u$ 是上界要求每个 $x \in A$ 满足 $x \preceq u$；确界还要求 $u$ 不大于任何其他上界。上界通常不在集合中。例如 $(0,1)$ 的确界是1，没有最大值。

# 10 原课件第 10 页：完备性预告




上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

非空有上界集合有确界是实数的关键性质，不是所有有序集合都满足。正有理数中平方小于2的数有上界，但在有理数内没有确界。

# 11 原课件第 11 页：区间



## Preorderings

Let $S$ be a preordered set. A subset $I$ of $S$ is an interval if "$x_1 \in I$, $x_2 \in I$ and $x_1 \precsim x \precsim x_2$" implies "$x \in I$." Let $a, b$ be two elements of $S$ such that $a \precsim b$; particular cases of intervals are:

$$[a,b] = \{x \in S \mid a \precsim x \precsim b\}, \qquad (a,b) = \{x \in S \mid a \prec x \prec b\},$$
$$[a,b) = \{x \in S \mid a \precsim x \prec b\}, \qquad (a,b] = \{x \in S \mid a \prec x \precsim b\},$$
$$[a,\rightarrow) = \{x \in S \mid a \precsim x\}, \qquad (\leftarrow,b] = \{x \in S \mid x \precsim b\}.$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

区间包含任意两点之间的全部可比较点。实数中 $[a,b]$ 包含端点，$(a,b)$ 不含端点。写邻域时括号变化会影响端点是否属于集合。

# 12 原课件第 12 页：乘积序



## Preorderings

Denote by $S_1, \cdots, S_n, \cdots, S_N$ $N$ preordered sets, by $\precsim_n$ the preordering on $S_n$, by $x_n$ a generic element of $S_n$. A preordering $\precsim$ is defined on the product $S = \prod_{n=1}^{N} S_n$ by $(x_n) \precsim (x'_n)$ if $x_n \precsim_n x'_n$ for every $n\,(= 1, 2, \cdots, N)$. According to the general notation of 1.4.c, $(x_n) \prec (x'_n)$ means that $(\alpha)$ for all $n$, $x_n \precsim_n x'_n$ and $(\beta)$ not for all $n$, $x'_n \precsim_n x_n$, i.e., $(\beta)$ for at least one $n$, $x_n \prec_n x'_n$. The notation $(x_n) \precnsim (x'_n)$ will express that, for all $n$, $x_n \prec_n x'_n$. With $n$ the exception of trivial cases, the preordering $\precsim$ on $S$ cannot be complete.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$x \le y$ 表示逐坐标不大于；$x < y$ 表示逐坐标不大于且至少一处严格；$x \ll y$ 则要求每处严格。例 $(0,1) < (1,1)$ 但不是 $\ll$。

# 13 原课件第 13 页：单调映射




Let $S$ and $T$ he two preordered sets, and denote by $\precsim_S$ (resp. $\precsim_T$) the preordering on $S$ (resp. on $T$). A function $f$ from $S$ to $T$ is said to be increasing (or to be a representation of $S$ in $T$) if "$x \sim_S x'$" implies "$f(x) \sim_T f(x')$" and "$x \prec_S x'$" implies "$f(x) \prec_T f(x')$".

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

序保持是 $x \preceq_S y \Rightarrow f(x) \preceq_T f(y)$。证明须从任意可比较的输入开始，不能只测几个值。严格递增、递减与弱递增的条件应分别写明。

# 14 原课件第 14 页：有序对的正式定义



## Binary Relations

For a given set $S$, let $S \times S$ denote the usual Cartesian product of all ordered pairs $(x_1, x_2)$, where both $x_1$ and $x_2$ are from $S$. A *binary relation* $\mathscr{R}$ on the set $S$ is formally defined as a subset of $S \times S$ — Write $\mathscr{R} \subseteq S \times S$, and $(x_1, x_2) \in \mathscr{R}$ if the ordered pair $(x_1, x_2)$ is in the relation $\mathscr{R}$. Another, quickly way to write $(x_1, x_2) \in \mathscr{R}$ is $x_1 \mathscr{R} x_2$, which can be read as "$x_1$ stands in the relation $\mathscr{R}$ to $x_2$." If $(x_1, x_2) \notin \mathscr{R}$, We'll write "$x_1 \not\mathscr{R} x_2$".

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

关系是集合，因此能取补、逆、交等。反关系把 $(x, y)$ 换成 $(y, x)$，补关系表示原判断不成立。以后“负传递”就是补关系的传递性。

# 15 原课件第 15 页：七种性质逐项读



## Binary Relations

A binary relation $\mathscr{R}$ on a set $S$ is:

- *reflexive* if $x\mathscr{R}x$ for all $x \in S$;
- *irreflexive* if $x\not\mathscr{R}x$ for all $x \in S$;
- *symmetric* if $x_1\mathscr{R}x_2$ implies $x_2\mathscr{R}x_1$;
- *asymmetric* if $x_1\mathscr{R}x_2$ implies $x_2\not\mathscr{R}x_1$;
- *antisymmetric* if $x_1\mathscr{R}x_2$ and $x_2\mathscr{R}x_1$ implies $x_1 = x_2$;
- *transitive* if $x_1\mathscr{R}x_2$ and $x_2\mathscr{R}x_3$ implies $x_1\mathscr{R}x_3$;
- *negatively transitive* if $x_1\not\mathscr{R}x_2$ and $x_2\not\mathscr{R}x_3$ implies $x_1\not\mathscr{R}x_3$;
- *complete* or *connected* if for all $x_1 \in S$ and $x_2 \in S$, $x_1\mathscr{R}x_2$ or $x_2\mathscr{R}x_1$ (or both);
- *weakly connected* if for all $x_1 \in$ and $x_2 \in S$, $x_1 = x_2$ or $x_1\mathscr{R}x_2$ or $x_2\mathscr{R}x_1$;
- *acyclic* if $x_1\mathscr{R}x_2, \cdots, x_{N-1}\mathscr{R}x_N$ imply $x_1 \neq x_N$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

自反与非自反不互为一般否定：否定自反只要求存在一个点不与自身有关。非对称推出反对称，但反对称允许对角点。做题把每个定义的全称量词写出来。

# 16 原课件第 16 页：严格偏好的要求



## Binary Relations

An agent is willing to express his preferences among these items by making paired comparisons of the form: "I strictly prefer $x_1$ to $x_2$" which is written $x_1 \succ x_2$. "Strict preference" is a binary relation on $S$. Consider the following properties that this binary relation might possess:

(1) *Asymmetry* - if $x_1 \mathscr{R} x_2$, then $x_2 \not\mathscr{R} x_1$.

(2) *Transitivity* - if $x_1 \mathscr{R} x_2$ and $x_2 \mathscr{R} x_3$, then $x_1 \mathscr{R} x_3$.

(3) *Irreflexivity* - $x \not\mathscr{R} x$ for all $x \in S$.

(4) *Negative transitivity* - if $x_1 \not\mathscr{R} x_2$ and $x_2 \not\mathscr{R} x_3$, then $x_1 \not\mathscr{R} x_3$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本课程用非对称与负传递刻画严格偏好。负传递是 $x \not\succ y,\ y \not\succ z \Rightarrow x \not\succ z$，不是普通传递换一个名字。

# 17 原课件第 17 页：负传递的等价形式



## Binary Relations

**[Lemma 1.]** A binary relation $\mathscr{R}$ is negatively transitive if and only if $x_1 \mathscr{R} x_2$ implies that, for all $x \in S$, $x_1 \mathscr{R} x$ or $x \mathscr{R} x_2$.

**Proof.**

$$
\begin{aligned}
& \text{Binary relation } \mathscr{R} \text{ is negatively transitive} \\
\iff & \text{If } x_1 \mathscr{P} x \text{ and } x \mathscr{P} x_2 \text{ for some } x \in S, \text{ then } x_1 \mathscr{P} x_2 \\
\iff & \text{If not } x_1 \mathscr{P} x_2, \text{ then not } [x_1 \mathscr{P} x \text{ and } x \mathscr{P} x_2] \text{ for all } x \in S \\
\iff & \text{If not } x_1 \mathscr{P} x_2, \text{ then not } x_1 \mathscr{P} x \text{ or not } x \mathscr{P} x_2 \text{ for all } x \in S \\
\iff & \text{If } x_1 \mathscr{R} x_2, \text{ then } x_1 \mathscr{R} x \text{ or } x \mathscr{R} x_2 \text{ for all } x \in S.
\end{aligned}
$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

原式的逆否命题是 $x \succ z \Rightarrow (x \succ y$ 或 $y \succ z)$，对每个中间点 $y$ 成立。推导用到 $\neg(P \wedge Q) = \neg P \vee \neg Q$，注意“或”允许两者同时成立。

# 18 原课件第 18 页：严格偏好定义



## Preference Relations

A binary relation $\succ$ on a set $S$ is called a strict preference relation if it is asymmetric and negatively transitive.

(1) $\succ$ *is asymmetric:* if $x_1 \succ x_2$, then $x_2 \not\succ x_1$.

(2) $\succ$ *is negative transitive:* if $x_1 \not\succ x_2$ and $x_2 \not\succ x_3$, then $x_1 \not\succ x_3$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

这是后面所有偏好推论的假设；不把它当作现实中自动满足的规律。出现循环选择就可能违反这些条件，数学模型是否适合现实需另行判断。

# 19 原课件第 19 页：传递与无循环的证明



## Preference Relations

**[Proposition 1.]** If $\succ$ is a strict preference relation, then it is irreflexive, transitive, and acyclic.

**Proof.** [1] Asymmetry directly implies irreflexivity.

[2] Suppose $x_1 \succ x_2$ and $x_2 \succ x_3$. By negative transitivity and Lemma 1, $x_1 \succ x_2$ implies either $x_1 \succ x_3$ or $x_3 \succ x_2$. But $x_3 \succ x_2$ is impossible because $x_2 \succ x_3$ is assumed and $\succ$ is asymmetric. Thus $x_1 \succ x_3$, which is transitivity.

Suppose $x_1 \succ x_2$ and $x_2 \succ x_3$. By negative transitivity and Lemma 1, $x_2 \succ x_3$ implies either $x_2 \succ x_1$ or $x_1 \succ x_3$. But $x_2 \succ x_1$ is impossible because $x_1 \succ x_2$ is assumed and $\succ$ is asymmetric. Thus $x_1 \succ x_3$, which is transitivity.

[3] If $x_1 \succ x_2, \cdots, x_{N-1} \succ x_N$, then by transitivity $x_1 \succ x_N$. Since $\succ$ is irreflexive, this implies $x_1 \neq x_N$. Thus $\succ$ is acyclic.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

非对称排除 $x \succ x$。若 $x \succ y \succ z$，由上页等价式，$x \succ y$ 推出 $x \succ z$ 或 $z \succ y$；后者与 $y \succ z$ 矛盾，故传递。沿有限循环反复传递得到 $x \succ x$，矛盾。

# 20 原课件第 20 页：从严格定义弱偏好



## Preference Relations

When we are given a binary relation $\succ$ that expresses strict preference, we use it to define two other binary relations:

$$x_1 \succsim x_2 \quad \text{if} \quad x_2 \not\succ x_1$$

and

$$x_1 \sim x_2 \quad \text{if} \quad x_1 \not\succ x_2 \quad \text{and} \quad x_2 \not\succ x_1$$

where we are using $\not\succ$ as shorthand for "not $\succ$". The relation $\succsim$ is called *preference*, although it really expresses the absence of strict preference. The preference $\sim$ is called *indifference* — it expresses the absence of strict preference in either direction, which is perhaps not quite the same thing as active indifference.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

定义 $x \succeq y \iff \neg(y \succ x)$，无差异表示两个严格方向都不成立。方向反转是易错点：弱偏好是“不严格差于”，不是“不严格好于”。

# 21 原课件第 21 页：三个互斥情形



## Preference Relations

**[Proposition 2.]** If $\succ$ is a strict preference relation, then

(1) For all $x_1$ and $x_2$, exactly one of $x_1 \succ x_2$, $x_2 \succ x_1$ or $x_1 \sim x_2$ holds.

(2) $\succsim$ is complete and transitive.

(3) $\sim$ is reflexive, symmetric, and transitive.

(4) $x_1 \succ x_2$, $x_2 \sim x_3$, $x_3 \succ x_4$ imply $x_1 \succ x_3$ and $x_2 \succ x_4$.

(5) $x_1 \succsim x_2$ if and only if $x_1 \succ x_2$ or $x_1 \sim x_2$.

(6) $x_1 \succsim x_2$ and $x_2 \succsim x_1$ imply $x_1 \sim x_2$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

非对称排除同时双向严格，余下恰为无差异。弱偏好传递由负传递得到：$\neg(y \succ x)$ 与 $\neg(z \succ y)$ 推出 $\neg(z \succ x)$。无差异的传递来自两条弱偏好链分别传递。

# 22 原课件第 22 页：命题证明展开



## Preference Relations

**Proof**. [1] follows from the definition of $\sim$ and the fact that $\succ$ is asymmetric.

[2] By the asymmetry of $\succ$, either $x_1 \not\succ x_2$ and $x_2 \not\succ x_1$ (or both) for all $x_1$ and $x_2$, thus $\succsim$ is complete. For the transitivity of $\succsim$, note that this follows immediately the negative transitivity of $\succ$.

[3] $\sim$ is reflexive because $\succ$ is irreflexive. $\sim$ is symmetric because the definition of $\sim$ is symmetric. For the transitivity, suppose $x_1 \sim x_2 \sim x_3$. Then $x_1 \not\succ x_2 \not\succ x_3$ and $x_3 \not\succ x_2 \not\succ x_1$. By negative transitivity of $\succ$, $x_1 \not\succ x_3 \not\succ x_1$, or $x_1 \sim x_3$.

[4] If $x_1 \succ x_2 \sim x_3$, then by [1] one of $x_1 \succ x_3$ or $x_3 \sim x_1$ or $x_1 \succ x_3$. But $x_3 \succ x_1$ is impossible, since then transitivity of $\succ$ would imply $x_3 \succ x_2$. And $x_3 \sim x_1$ is impossible, since then transitivity of $\sim$ would imply $x_2 \sim x_1$, contradicting $x_1 \succ x_2$. Thus $x_1 \succ x_3$ must hold.

If $x_2 \sim x_3 \succ x_4$, then by [1] one of $x_2 \succ x_4$ or $x_4 \sim x_2$ or $x_2 \succ x_4$. But $x_4 \succ x_2$ is impossible, since then transitivity of $\succ$ would imply $x_3 \succ x_2$. And $x_4 \sim x_2$ is impossible, since then transitivity of $\sim$ would imply $x_4 \sim x_3$, contradicting $x_3 \succ x_4$. Thus $x_2 \succ x_4$ must hold.

[5] $x_1 \succsim x_2$ if and only if $x_2 \not\succ x_1$ if and only if $x_1 \succ x_2$ or $x_1 \sim x_2$ (by part [1]).

[6] This is immediate from the definition of $\succsim$ and $\sim$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若 $x \succ y$ 且 $y \sim z$，由负传递等价式 $x \succ z$ 或 $z \succ y$，后一项被无差异排除。若还 $z \succ w$，再用传递得 $x \succ w$。每个被排除的分支都应写出所违反的假设。

# 23 原课件第 23 页：两种原始描述



## Preference Relations

Note well the plot: the agent expresses strict preferences, from which *we* define preferences and indifference. It is strict preference that is basic — the agent is not being called upon to express any judgments concerning weak preference or indifference, and he might disagree with our use of those terms to describe the negation of strict preference.

Another possible plot would be to ask the agent to express preferences or strict preference or indifference. That is, the basic relation is $\succsim$. That is a plot that is followed in many developments of choice theory, and in the standard treatment it leads to the same mathematical results.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页解释以严格偏好为基础的建模选择。弱偏好和无差异是用定义导出的，不应再任意赋值，否则可能互相矛盾。

# 24 原课件第 24 页：弱偏好定义



## Preference Relations

A binary relation $\succsim$ on a set $S$ is called a preference relation if it is complete and transitive.

(1) $\succsim$ *is complete:* for all $x_1 \in S$ and $x_2 \in S$, $x_1 \succsim x_2$ or $x_2 \succsim x_1$ (or both).

(2) $\succsim$ *is transitive:* if $x_1 \succsim x_2$ and $x_2 \succsim x_3$, then $x_1 \succsim x_3$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

完全性包括与自身比较，因而推出自反。传递要求任何三点的两段比较能够连接。没有完全性时，$\neg(y \succeq x)$ 不足以推出 $x \succeq y$。

# 25 原课件第 25 页：反向构造



## Preference Relations

When we are given a binary relation $\succsim$ that expresses preference, we use it to define two other binary relations:

$$x_1 \succ x_2 \quad \text{if} \quad x_2 \not\succsim x_1$$

and

$$x_1 \sim x_2 \quad \text{if} \quad x_1 \succsim x_2 \quad \text{and} \quad x_2 \succsim x_1$$

where we are using $\not\succsim$ as shorthand for "not $\succsim$". The relation $\succ$ is called *strict preference*, although it really expresses the absence of preference. The preference $\sim$ is called *indifference* — it expresses the preference in both directions.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

给定完全传递的 $\succeq$，定义严格关系为弱关系成立且反向不成立；完全性使其等价于单写反向不成立。无差异仍为双向弱关系。

# 26 原课件第 26 页：构造的性质



## Preference Relations

**[Proposition 3.]** If $\succsim$ is a preference relation, then

(1) For all $x_1$ and $x_2$, $x_1 \succsim x_2$ or $x_2 \succsim x_1$ (or both) holds.

(2) $\succ$ is asymmetric and negatively transitive.

(3) $\sim$ is reflexive, symmetric, and transitive.

(4) $x_1 \succ x_2$ if and only if $x_1 \succsim x_2$ but $x_2 \not\succsim x_1$.

(5) $x_1 \succ x_2$ if and only if $x_1 \succsim x_2$ but $x_2 \nsim x_1$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

严格关系非对称：若 $x \succ y$，则 $x \succeq y$，故不能同时 $y \succ x$。负传递：$x \not\succ y$ 意味 $y \succeq x$，再与 $z \succeq y$ 传递即得 $x \not\succ z$。无差异为等价关系，由自反、交换方向和双链传递得到。

# 27 原课件第 27 页：原页无差异公式纠错



## Preference Relations

**[Proposition 4.]** Given a preference relation $\succsim'$ on a set $S$, define two new binary relation $\succ'$ and $\sim'$ from $\succsim'$ by

$$x_1 \succ' x_2 \quad \text{if} \quad x_2 \not\succsim' x_1$$

and

$$x_1 \sim' x_2 \quad \text{if} \quad x_1 \succsim' x_2 \quad \text{and} \quad x_2 \succsim' x_2.$$

Then if $\succsim'$ is complete and transitive, $\succ'$ will be a strict preference relation. Moreover, if we start with a binary relation $\succsim'$, define $\succ'$ and $\sim'$ as above from $\succsim'$, and then define $\succsim$ and $\sim$ from $\succ'$ by

$$x_1 \succsim x_2 \quad \text{if} \quad x_2 \not\succ' x_1$$

and

$$x_1 \sim x_2 \quad \text{if} \quad x_1 \not\succ' x_2 \quad \text{and} \quad x_2 \not\succ' x_1$$

then $\succsim'$ and $\succsim$ will agree, as will $\sim'$ and $\sim$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

原图第二项应为 $x_2 \succeq' x_1$，不能是 $x_2 \succeq' x_2$。后者总真，会把无差异误写成单向弱偏好。修正后上下两种出发方式互相还原。

# 28 原课件第 28 页：还原弱关系



## Preference Relations

**Proof.** $\succ'$ is asymmetric because $x_1 \succ' x_2$ implies $x_2 \not\succsim' x_1$ from the definition of $\succ'$ and then $x_2 \not\succ' x_1$. $\succ'$ is negatively transitive because $x_1 \not\succ' x_2$ and $x_2 \not\succ' x_3$ implies $x_3 \succsim' x_2$ and $x_2 \succsim' x_1$ and then $x_3 \succsim' x_1$ from the completeness and transitivity of $\succsim'$, thus $x_1 \not\succ' x_3$ from the definition of $\succ'$.

$x_1 \succsim' x_2$ implies $x_2 \not\succ' x_1$ from the completeness of $\succsim'$, then $x_1 \succsim x_2$ from the definition of $\succsim$. $x_1 \succsim x_2$ implies $x_2 \not\succ' x_1$ from the definition of $\succsim$, then $x_1 \succsim' x_2$ from the completeness of $\succsim'$.

The proof of that $\sim'$ and $\sim$ are equivalent is from the definitions of the indifferences.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

原构造给出 $x \succeq y \iff \neg(y \succ' x) \iff \neg\neg(x \succeq' y) \iff x \succeq' y$。严格关系和无差异也由相同定义还原。这里的双重否定是逻辑步骤，不是经验判断。

# 29 原课件第 29 页：本讲复习



## Preference Relations

So it doesn't matter whether we start with a strict preference relation that is asymmetric and negatively transitive or with a preference relation that is complete and transitive — we end up to in the same place. For reasons of interpretation we prefer to take strict preference as being basic. But it is a matter of personal taste, and most authors do it the other way.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

分别从严格和弱偏好出发重写定义，检查最终回到原关系。自测：用效用 $u$ 定义 $x \succeq y \iff u(x) \geq u(y)$；实数顺序的完全和传递立即给出完全预序，但 $u(x) = u(y)$ 不推出 $x = y$，所以一般不是偏序。